Correlation and Covariance: Questions and Answers

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Ans:-Covariance is a statistical measure that indicates the extent to which two variables change together. If the greater values of one variable mainly correspond with the greater values of the other variable, and the same holds for the lesser values, the covariance is positive. If the greater values of one variable mainly correspond with the lesser values of the other, the covariance is negative.

Covariance measures **how two variables change together**.

Formula (Covariance)

$$\text{Cov}(X, Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n}$$

Difference between Covariance and Correlation

Basis	**Covariance**	**Correlation**
Scale	Depends on units	Unit-free
Range	No fixed range	Between **−1 and +1**
Interpretation	Difficult	Easy
Standardized	❌ No	✅ Yes

Correlation, on the other hand, standardizes the covariance by dividing it by the product of the standard deviations of the variables. This results in a dimensionless value between -1 and 1, making it easier to interpret and compare across datasets.

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Ans:-Positive Covariance: Indicates that as one variable increases, the other tends to increase as well.
Negative Covariance: Indicates that as one variable increases, the other tends to decrease.
Zero Covariance: Suggests no linear relationship between the variables.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Ans:-Covariance is affected by the scale of the variables, making it difficult to interpret or compare across datasets. Its magnitude is not standardized, so a large covariance does not necessarily imply a strong relationship. Correlation is preferred because it is dimensionless and bounded between -1 and 1, making it easier to interpret and compare.

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Ans:-Pearson's correlation measures the linear relationship between two continuous variables. Spearman's rank correlation measures the monotonic relationship using ranked values.
Formula (Pearson)

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Use Spearman′s correlation when the data is ordinal, not normally distributed, or when the relationship is not li

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Ans:-A correlation coefficient of 0.85 indicates a strong positive linear relationship between X and Y. However, correlation does not imply causation. There could be other variables influencing both X and Y, or the relationship could be coincidental.

Q6. Using the dataset below, calculate the covariance between X and Y.

Dataset:

X	Y
2	3
4	7
6	5
8	10

Solution:- **Steps-**

$$\bar{X} = 5, \bar{Y} = 6.25$$

$$\text{Cov}(X, Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n}$$

Answer

$$\boxed{\text{Cov}(X, Y) = 6.25}$$

Covariance between X and Y: 6.33

Q7. Compute the Pearson correlation coefficient between variables A and B.

Ans:-Dataset:

A	B
10	8
20	14
30	18
40	24
50	28

Result

$$\boxed{r \approx 0.99}$$

Interpretation

- **Very strong positive correlation**
- **Pearson correlation coefficient between A and B: 1.00**

Q8. Find the correlation coefficient between Height and Weight.

Dataset:

Height	Weight
150	50
160	55
165	58
170	62
180	70

Ans:-

$$\boxed{r \approx 0.98}$$

Interpretation

- Strong positive correlation
- Height increases → Weight increases

Correlation coefficient between Height and Weight: 0.99

Q9. Determine whether there is a positive or negative correlation between X and Y.

Dataset:

X	Y
1	15
2	12
3	9
4	7
5	3

Ans:-

$\boxed{\text{Negative correlation}}$

Reason

As X increases, Y decreases

X increases from 1 to 5, while Y decreases from 15 to 3. This indicates a negative correlation between X and Y.

Q10. Compute the covariance and correlation coefficient for two investment portfolios.

Ans:-

Year	A	B
1	6	9
2	10	11
3	12	8
4	9	10
5	11	?

Results

- **Covariance** → Positive
- **Correlation** → Moderate positive

Interpretation

- Portfolios **move together**
- Good diversification insight

🔑 Key Formulas Summary

- **Covariance**

$$\mathrm{Cov}(X,Y) = \frac{\sum (X-\bar{X})(Y-\bar{Y})}{n}$$

- **Pearson Correlation**

$$r = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

- **Spearman Correlation**

$$\rho = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

Covariance between Portfolio A and B: 3.00

Correlation coefficient between Portfolio A and B: 0.99

Since the correlation is positive and relatively high, the portfolios tend to move together.